
DAY 857
of 2,451 days
Gravity assists (Cassini path)
NEXT: JUPITER FLYBY
SPACE HOLD TO FAST-FORWARD
EARTH FLYBY: FREE SPEED FROM GRAVITY
SHIP CAM
ASTROPIXEL
MISSION DESIGNER
URANUS
EARTH
MERCURY
NEPTUNE
SATURN
Design a spacecraft, launch it and fly the whole mission,
as a team of up to 4, with real NASA numbers.
When it fails, the game tells you why.
Team Shay Rafe3 · Challenge: Space Mission Design Game · Ramallah local event
Omar · Ahmad · Ameen · Mahdi
LIFT
CRUISE
FUEL
191 kg
BATT
none
Cruise: 2451 days. Power OK (330 W made, 180 W used). Course corrections used 8.8 kg of fuel.

IMPACT

Most student satellites never finish the mission

1 in 4

university satellites finished their full mission

244 university smallsats launched 2000–2015. Nearly 30% were dead on arrival.

3 in 5

satellites lost by teams on their first 1–3 tries

1 in 3

CubeSat failures with no known cause

Source: Swartwout (Saint Louis Univ.) and Langer (TU Munich), in The Space Review, 22 Aug 2016

Tools teach one topic

Orbits, rocket design or launches on their own. Nobody shows how the parts affect each other.

Failure without a reason

A mission fails on mass, power, data or one missing part. Without the why, it just feels random.

Real missions are teamwork

NASA designs missions as a team of specialists who must agree. Most learning games are solo.

CREATIVITY RELEVANCE

One game. The whole mission. A real team.



The hangar: parts snap onto the ship; Mission Control lamps turn green only when NASA's margins are kept.

8
worlds × 3 tasks

28
real parts

4
player roles

37
real-mission events

What's new

- Unlike Kerbal Space Program, every choice costs budget, power, data, comms and time, and every number has a NASA source.
- Unlike NASA's Webb game (one telescope, solo), you fly 8 destinations as a team of 4 who must sign off together.

RELEVANCE VALIDITY

NASA data drives every result

NASA DATA

DONKI
2024 geomagnetic storms (GST), up to Kp 9

NSSDCA Planetary Fact Sheets
gravity, sunlight, distance

JPL Concept Maturity Level 4
35% margins, 3 dB link

GMAT
engine Isp 300 s, solar ageing

MMRTG · Small Spacecraft Tech
real power parts

GAME RULES

Rocket equation
heavier ship → more fuel → heavier ship

Sunlight ∝ 1 / distance²
Saturn gets ~1% of Earth's

Radio ∝ 1 / distance²
far worlds need a big dish

Margins
lamp is green only with NASA's margin

Events eat margin
no margin → first surprise ends it

WHAT THE PLAYER LIVES

A real storm in cruise
no shielding → 70% chance a part fails

Solar ship at Saturn
goes dark: needs nuclear power

Mars landing
heat shield + parachute, real gravity

The report
4 scores + the cause chain: why

Redesign
and fly again

$\Delta v = Isp \cdot g_0 \cdot \ln(m_0 / m_1)$ Every number lives in one file with its source. Values we chose for gameplay are labelled "game rule".

VALIDITY PRESENTATION

Play it now, in any browser



1 Fly the launch: throttle back through max-Q, stage, deploy



2 Decide: events from real missions, each role owns some



3 Land in real gravity: Venus, 464 °C and 92 bar

Scan to play
free · no install · no login

WORKS TODAY

- All 8 worlds × 3 tasks playable start to finish
- Free web build on GitHub Pages: no install, no login
- Multiplayer for 1–4 roles, tested in 2 browsers on a local server

PLANNED

- Public relay server so judges can join online games
- Full GMAT trajectory runs instead of real-mission Δv
- Real NASA photos (LRO, HiRISE, Cassini) in the report

IMPACT

Who it helps, and how



Students and new engineers

Learn mass, power, data and comms by using them, in plain words, from zero knowledge up.



Teachers and schools

One free link for a whole class: 4 roles per team, a report that explains every result.



Space clubs and hackathons

Team play over the network: negotiate trade-offs like a real NASA design team.

LOCAL ANGLE

Runs in any browser on a school PC or a phone-tethered laptop: no install, no account, no cost.

Built in Ramallah, for students who rarely get hands-on space engineering.

TESTED, THEN REBUILT

Playtest of our early version: the builder's list of +/- counters was "too complicated".

So we rebuilt it: click a slot on the ship, pick a real part, and a hint names the part that fixes the problem.

VALIDITY IMPACT

From a 48-hour build to a classroom

1. **Deploy the relay server**
   so anyone can play online as a team
2. **Pilot with one class**
   and fix what first-timers trip on
3. **Full GMAT runs + NASA photos**
   for exact flight paths and real images

HONEST LIMITS

- Robotic missions only: no crew, no life support
- Flight times and Δv from real missions, not full GMAT runs
- Online play needs our relay server

THE TEAM · SHAY RAFE3 · RAMALLAH

Omar
Ahmad
Ameen
Mahdi

Play: https://omarshobaki-png.github.io/Shay-Rafe3-Nasa-Space-App-Challenge/play/ · AI disclosure: the idea and game design are ours; AI helped write code and draft this deck. No AI images, video or audio.